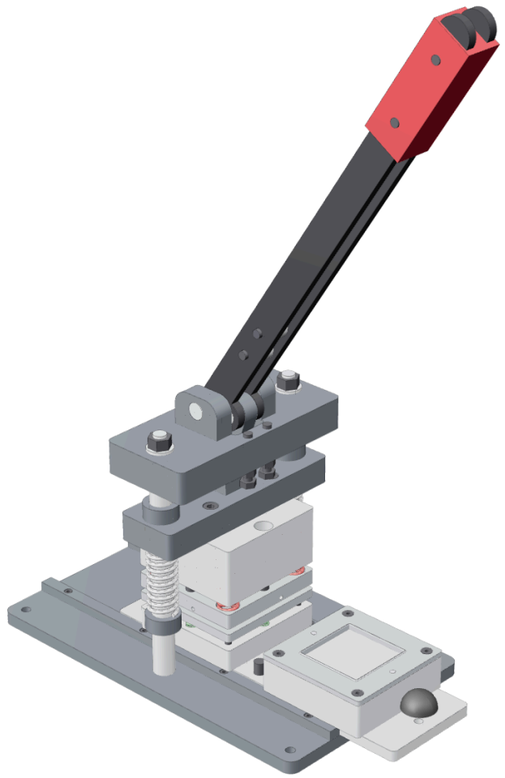
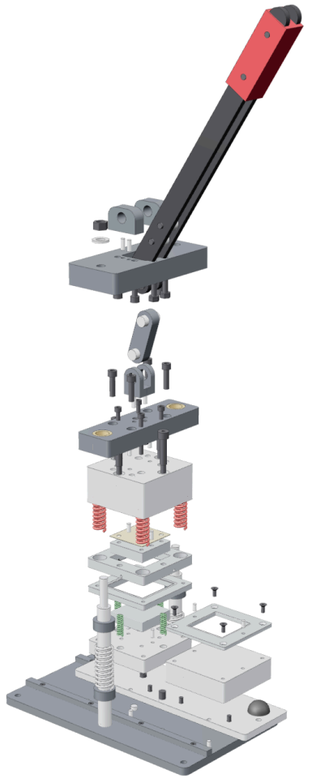
MagPress L1

Light photo-magnet press, 63.5 x 63.5 (2.5 inch square), market edge





Footprint	360 x 220 mm (base), height about 652 mm with the handle up
Mass	about 26 kg
Stroke	14.1 mm (handle 57 deg up to -2.5 deg down)
Force at the die	about 2 kN fold (stroke 1), 3-5 kN with the upper springs (stroke 2); frame checked at 15 kN
Magnet	63.5 x 63.5, finished about 4.4 thick

SHEET LIST

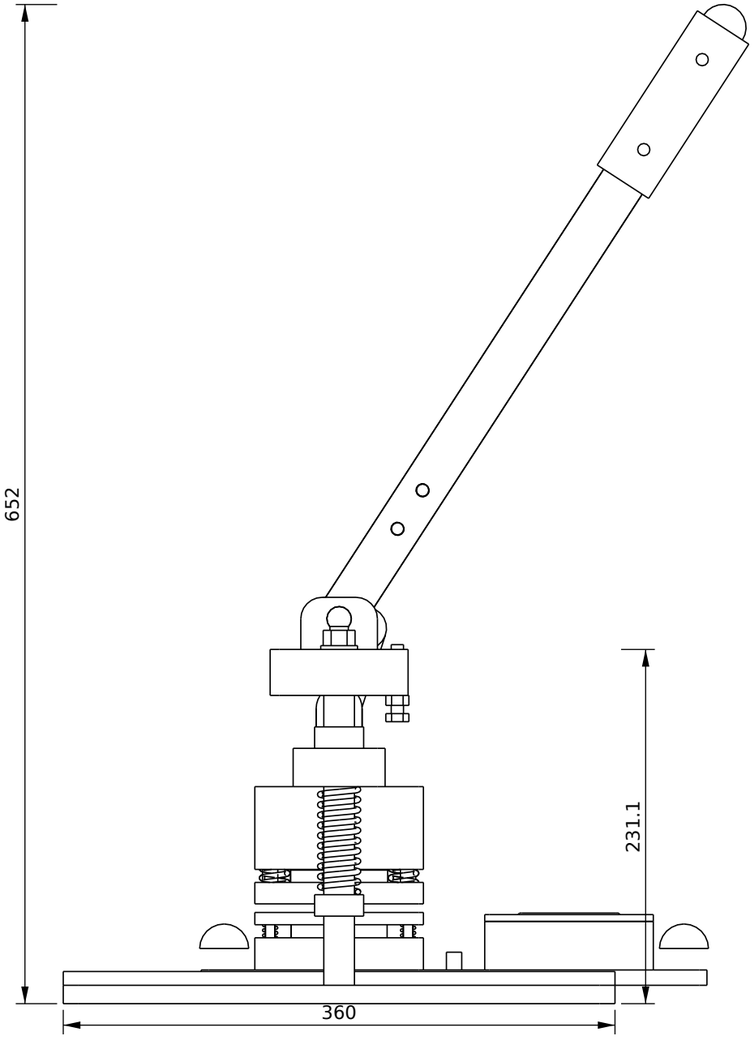
SHEET	DRAWING	TITLE	QTY
1	L-000	COVER AND SHEET LIST	-
2	L-001	GENERAL ARRANGEMENT	1
3	L-002	DIE SECTIONS AT THE BOTTOM OF THE STROKE	-
4	L-003	MAGNET EDGE AND MATERIAL (ASSUMED)	-
5	L-004	ASSEMBLY SEQUENCE AND USE	-
6	L-101	BASE PLATE	1
7	L-102	TABLE RAIL	2
8	L-103	SLIDING TABLE	1
9	L-104	COLUMN	2
10	L-105	HEAD	1
11	L-106	RAM	1
12	L-107	PIVOT CHEEK	2
13	L-108	RAM CLEVIS	1
14	L-109	LEVER PLATE	2
15	L-110	TOGGLE LINK	1
16	L-111	PIN O (STUB)	2
17	L-112	PIN A / PIN B	2
18	L-113	LEVER SPACER TUBE	2
19	L-114	GRIP SPACER	1
20	L-201	STATION A BASE BLOCK	1
21	L-202	PEDESTAL	1
22	L-203	RING A (BLANK HOLDER)	1
23	L-204	STATION B RISER	1
24	L-205	CUP RING (TAPER + NOSE)	1
25	L-206	UPPER DIE BODY	1
26	L-207	UPPER RING	1
27	L-208	PLUG	1
28	L-209	PLUG SHIM	2 x 0.1, 1 x 0.2

NOTES

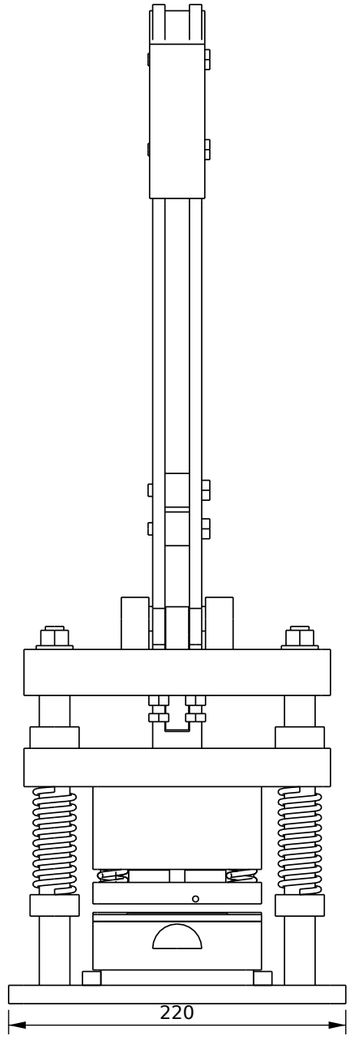
1. MagPress L1: a light, hand-lever photo-magnet press for one size, 63.5 x 63.5 (2.5 inch square) with the market edge: the flat tin is folded once (sharp crease, flap angled back onto the acrylic card) and the photo and film wrap round it.
2. Two stations on a sliding table: A folds the flat tin with the photo and film into a shell (stroke 1); B presses the shell onto the acrylic card and magnet sheet and turns the flap in (stroke 2). One magnet = 2 strokes.
3. ASSUMED material (to confirm with a seller's pre-cut set before manufacture): tin 71.5 x 71.5 x 0.23, photo and film 79 x 79 octagons, acrylic card 60.5 x 60.5 x 3.0, magnet sheet 57 x 57 x 1.0. Parts marked MAT follow these sizes.
4. DXF files: one per cut part, 1:1 mm, layers OUTLINE / HOLES / TAP_DRILL / COUNTERBORE / POCKET. No part is bent: all are flat profiles (laser, plasma or wire-EDM) plus machining; columns and pins are turned.
5. 3D: STEP of every part and of the assembly come with this set.

PROJECT: MagPress L1 · light photo-magnet press	REVISION: **L1-A**	SHEET: 1 / 28		
TITLE: **COVER AND SHEET LIST**				
DRAWING NO.: **L-000**	QTY: -	SCALE: -	UNITS: mm	
MATERIAL: -				
HEAT TREATMENT / FINISH: -				
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-29	DRAWN:	CHECKED:

FRONT VIEW



SIDE VIEW (from the right)



PARTS LIST - MADE

PART	NAME	QTY	MATERIAL	KG
L-101	BASE PLATE	1	S275	7.39
L-102	TABLE RAIL	2	Bright mild steel flat 12 x 9	0.30
L-103	SLIDING TABLE	1	6061-T6 aluminium plate 10	0.84
L-104	COLUMN	2	CK45 / C45 induction-hardened hard-chrome ba	0.52
L-105	HEAD	1	S275	3.63
L-106	RAM	1	S275	1.99
L-107	PIVOT CHEEK	2	S275	0.18
L-108	RAM CLEVIS	1	C45 / EN8 block 35 x 35 x 45	0.15
L-109	LEVER PLATE	2	S275	0.90
L-110	TOGGLE LINK	1	S355	0.21
L-111	PIN O (STUB)	2	EN8 / C45 bright bar	0.04
L-112	PIN A / PIN B	2	EN8 / C45 bright bar	0.05
L-113	LEVER SPACER TUBE	2	EN8 / mild steel round bar D22	0.04
L-114	GRIP SPACER	1	S275	0.40
L-201	STATION A BASE BLOCK	1	6061-T6 aluminium plate 16, faced to 21	0.64
L-202	PEDESTAL	1	D2 / 1.2379 tool steel	0.44
L-203	RING A (BLANK HOLDER)	1	D2 / 1.2379 tool steel	0.43
L-204	STATION B RISER	1	6061-T6 aluminium plate 25, faced to 31.675	1.03
L-205	CUP RING (TAPER + NOSE)	1	D2 / 1.2379 tool steel	0.27
L-206	UPPER DIE BODY	1	6061-T6 aluminium plate 40, faced to 54	1.46
L-207	UPPER RING	1	D2 / 1.2379 tool steel	0.76
L-208	PLUG	1	D2 / 1.2379 tool steel	0.46
L-209	PLUG SHIM	2 x 0.1, 1 x 0.2	Stainless shim stock 0.1 / 0.2	0.01

PARTS LIST - BOUGHT

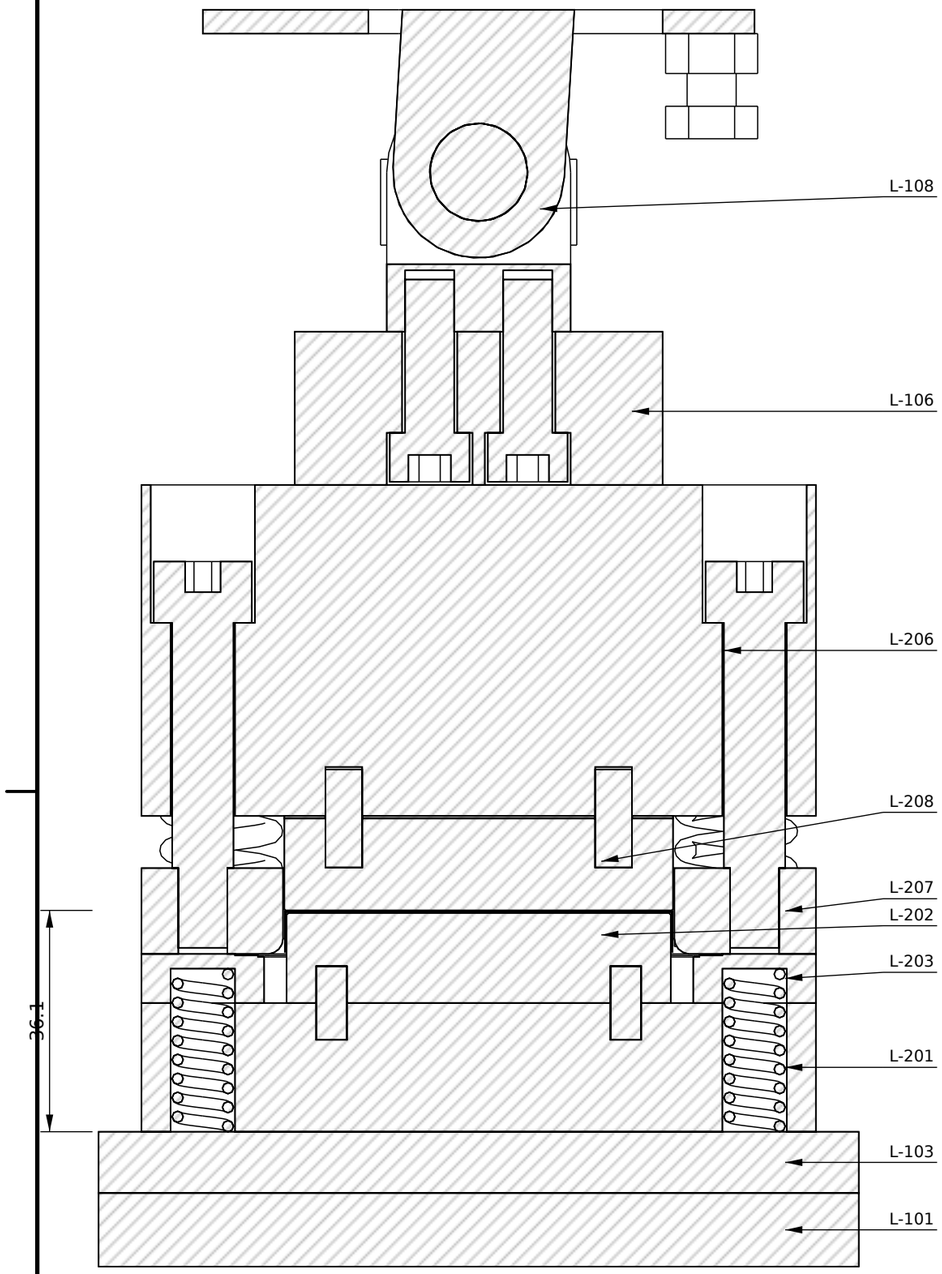
ITEM	DESCRIPTION	QTY	WHERE
B-01	Bronze plain bush D20 x D28 x 25 (or sintered bronze, oil-impregnated)	2	Ram on the columns
B-02	Hex nut M12 + washer M12	2 + 2	Columns over the head
B-03	Shaft collar DIN 705 A 20 (D32 x 14) with set screw	4	2 return-spring seats at Z 57.1, 2 up-stops at Z 166
B-04	Compression spring ID 21, wire 3.6, L0 80, 8 active coils, about 14 N/mm	2	Ram return, on the columns
B-06	Cap screw ISO 4762 M8 x 35 / M8 x 25	4 + 2	Head to cheeks; ram to clevis
B-07	Cap screw ISO 4762 M8 x 30	4	Ram to die body
B-08	Hex bolt M8 x 45 + hex nut M8	2	Handle stops in the head (one under each lever plate
B-08a	Hex bolt M8 x 35 + nyloc nut M8	4	Lever plates through the spacer tubes and the grip s
B-09	Dowel pin ISO 8734 D6 m6 x 16	15	Head/cheeks 4, ram/clevis 2, ram/die body 2, die bod
B-10	Dowel pin ISO 8734 D5 m6 x 12 / x 10	2 + 2	Pedestal; cup ring
B-11	Dowel pin ISO 8734 D10 m6 x 16	1	Table stop pin in the base
B-12	Ball plunger M10 (spring, ball D6, end force 30-60 N)	1	Table detent
B-13	Knob DIN 319 D32 M8	2	Table
B-14	Cap screw ISO 4762 M4 x 10	8	Rails
B-15	Cap screw ISO 4762 M6 x 12	8	Stations to table
B-16	Cap screw ISO 4762 M5 x 16	4	Pedestal
B-17	Shoulder screw ISO 7379 D6 x 16, M5 (thread cut to 7.5)	4	Ring A
B-18	Die spring ISO 10243 green D10 x 38	4	Ring A
B-19	Countersunk screw ISO 10642 M5 x 12	4	Cup ring
B-20	Shoulder screw ISO 7379 D10 x 40, M8	4	Upper ring
B-21	Die spring ISO 10243 red D20 x 76	4	Upper ring
B-22	Cap screw ISO 4762 M6 x 55	4	Plug through the die body and shims
B-23	Ball plunger M4 x 12 + set screw M4 x 6	4 + 4	Upper ring: hold the shell between strokes
B-24	Handle grip: rubber sleeve over the 32 x 30 grip end (or cloth tape)	1	Lever
B-25	Rubber feet / bench bolts M8	4	Base

NOTES

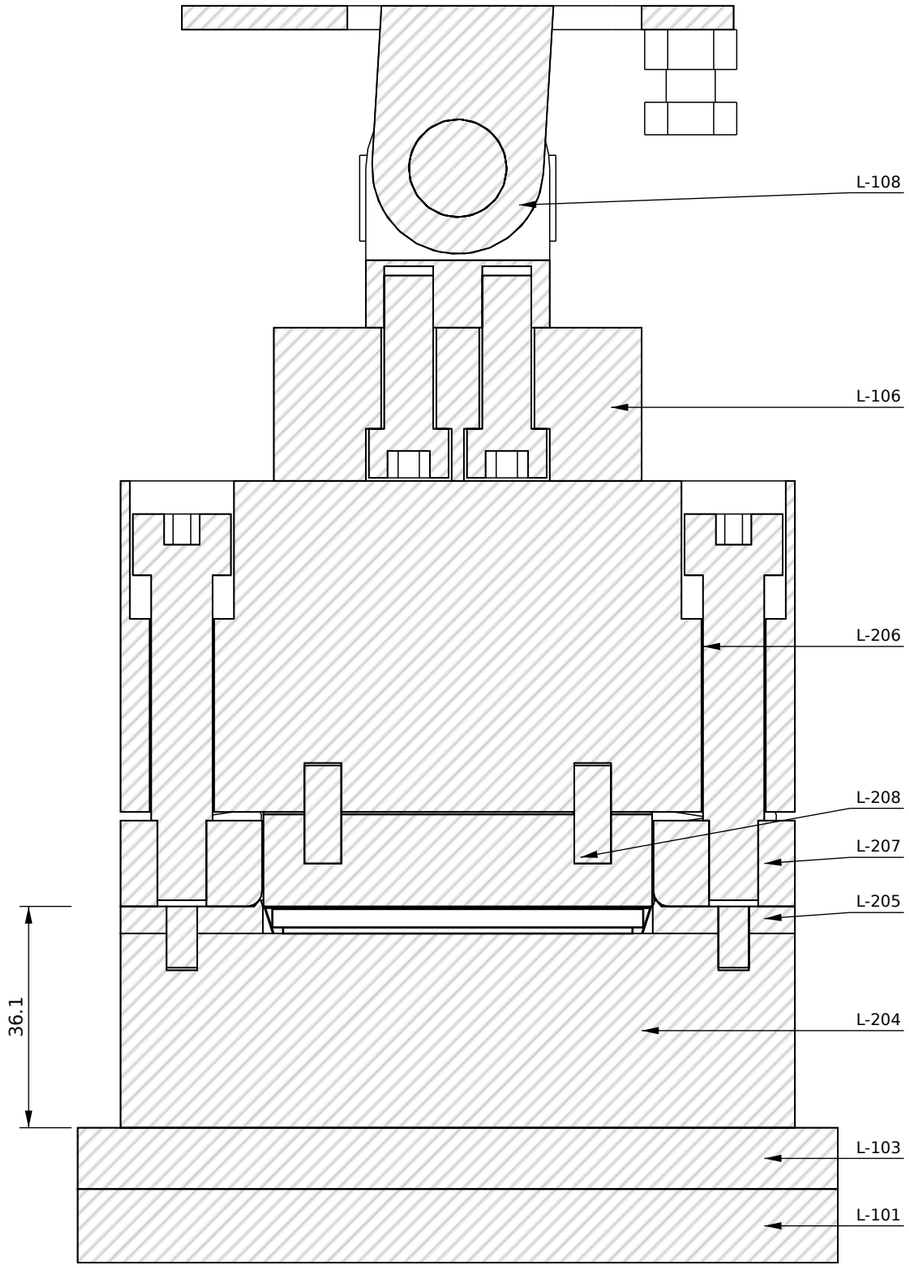
1. Handle up, station A (fold) under the ram. Pull the handle down to the stop for each stroke; slide the table between the two click positions (ball detent) to change station.
2. Heights at the bottom of the stroke: table top Z 22, magnet face 58.1, plug face 58.1. Pivot O at Z 251.1.
3. Bolt the base to a bench (4 x M8) or fit rubber feet. Grease the pins, the columns and the table slide.
4. Interference check (3D): no clashes through the stroke at either station or while the table slides (springs excluded).

PROJECT: MagPress L1 · light photo-magnet press		REVISION: **L1-A**	SHEET: 2 / 28
TITLE: **GENERAL ARRANGEMENT**			
DRAWING NO.: **L-001**	QTY: 1	SCALE: 1:4	UNITS: mm
MATERIAL: see parts list			
HEAT TREATMENT / FINISH: -			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-29	DRAWN / CHECKED

STATION A - STROKE 1



STATION B - STROKE 2



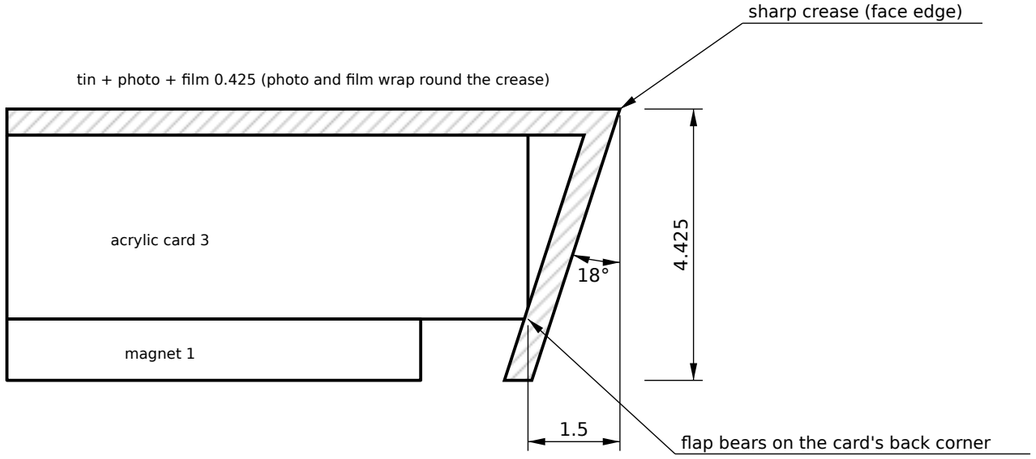
KEY

PART	NAME
L-101	base plate
L-103	sliding table
L-106	ram
L-108	ram clevis
L-201	station A base block
L-202	pedestal
L-203	ring A (blank holder)
L-204	station B riser
L-205	cup ring (18 deg taper)
L-206	upper die body
L-207	upper ring
L-208	plug
black	tin + photo + film
white	acrylic card, magnet sheet

NOTES

1. Sections on the press centre plane (Y = 0), handle on its stop, material shown black. Left: station A, stroke 1 (the flat tin, photo and film folded over the pedestal into the upper ring). Right: station B, stroke 2 (the shell pushed into the taper cup onto the acrylic card and magnet; the upper ring retracted 7.1 mm).
2. Both strokes end with the plug face at Z 58.1: the handle stop sets it (two stop bolts; 1 deg = 0.08 mm); the plug shims set station B against station A.
3. Stroke 1: ring A (4 light springs) carries the blank down 8.5 mm round the fixed pedestal; the upper ring (4 heavy springs, 2.9 kN preload) holds the edge, so the flat tin folds up the ring bore. Ball plungers keep the shell in the upper ring when the ram goes up.
4. Stroke 2: the upper ring lands on the cup ring, its mouth over the cup ring's nose; the plug pushes the shell down 7.1 mm and the 18 deg taper turns the tin flap in onto the back corner of the acrylic card.

PROJECT: MagPress L1 · light photo-magnet press		REVISION: **L1-A**	SHEET: 3 / 28
TITLE: **DIE SECTIONS AT THE BOTTOM OF THE STROKE**			
DRAWING NO.: **L-002**	QTY: -	SCALE: 1:1	UNITS: mm
MATERIAL: -			
HEAT TREATMENT / FINISH: -			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-29	DRAWN / CHECKED



MATERIAL PER MAGNET (ASSUMED - TO CONFIRM)

ITEM	SIZE (mm)	SOURCE
Tin sheet (flat)	71.5 x 71.5 x 0.23, corners clipped 45 deg	seller pre-cut
Photo	79 x 79 octagon, paper about 0.12	own print
Film	79 x 79 octagon, about 0.075	seller pre-cut
Acrylic card	60.5 x 60.5 x 3	seller pre-cut
Magnet sheet	57 x 57 x 1, self-adhesive, stuck centred on the card first	seller pre-cut
Finished magnet	63.5 x 63.5 face, about 4.42 thick	

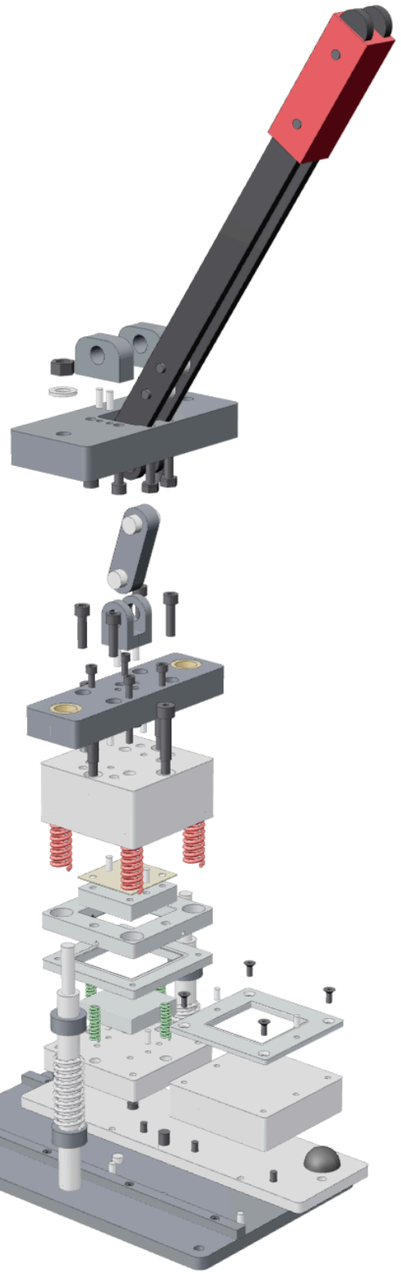
PARTS THAT CHANGE WITH THE MATERIAL (MAT)

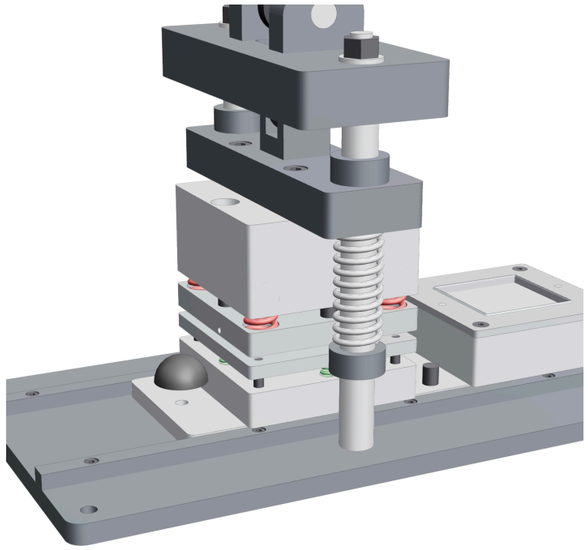
PART	DEPENDS ON
L-202 pedestal	face - 2 x (tin + photo + film)
L-203 ring A	pockets = tin and photo sizes, opening = tin - 1.5
L-205 cup ring	landing face height = finished magnet; taper; nose
L-204 riser	thickness = 28 - cup ring
L-207 upper ring	bore = face + 0.35
L-208 plug	face - 0.1

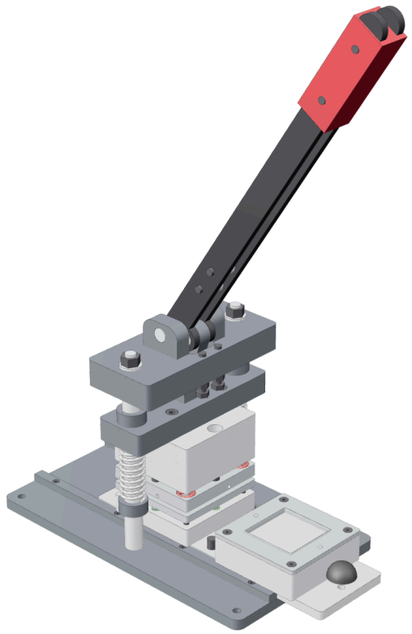
NOTES

1. The finished edge (market construction): tin folded once with a sharp crease at the face, flap angled back 18 deg and bearing on the back corner of the acrylic card; photo and film wrap round the crease and are tucked back between the flap and the card.
2. Sizes shown are ASSUMED. Measure one pre-cut set from the chosen seller (ruler accuracy) and send: tin L x W x t; photo / film L x W and corner clip; acrylic L x W x t; magnet L x W x t. Parts marked MAT are then adjusted.
3. Stroke 2 forming is checked in 2D (plane strain, verify/stroke2L.py); the paper tuck is not modelled. Prove it on the first die set with real material before ordering spares.

PROJECT: MagPress L1 · light photo-magnet press		REVISION: **L1-A**	SHEET: 4 / 28
TITLE: **MAGNET EDGE AND MATERIAL (ASSUMED)**			
DRAWING NO.: **L-003**	QTY: -	SCALE: 10:1	UNITS: mm
MATERIAL: -			
HEAT TREATMENT / FINISH: -			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-29	DRAWN / CHECKED







OPERATING CYCLE (one magnet = 2 strokes)

#	HANDLE / TABLE	WHAT HAPPENS
1	Table right (station A under the ram)	load B: card + magnet into the cup
2	Pull to the stop	stroke 1 at A: flat tin + photo + film folded into a shell
3	Slide the table left (B under the ram)	load A: next tin, photo, film
4	Pull to the stop	stroke 2 at B: shell pressed onto the card, flap turned in
5	Slide right, lift the magnet out	repeat from 2

ASSEMBLY

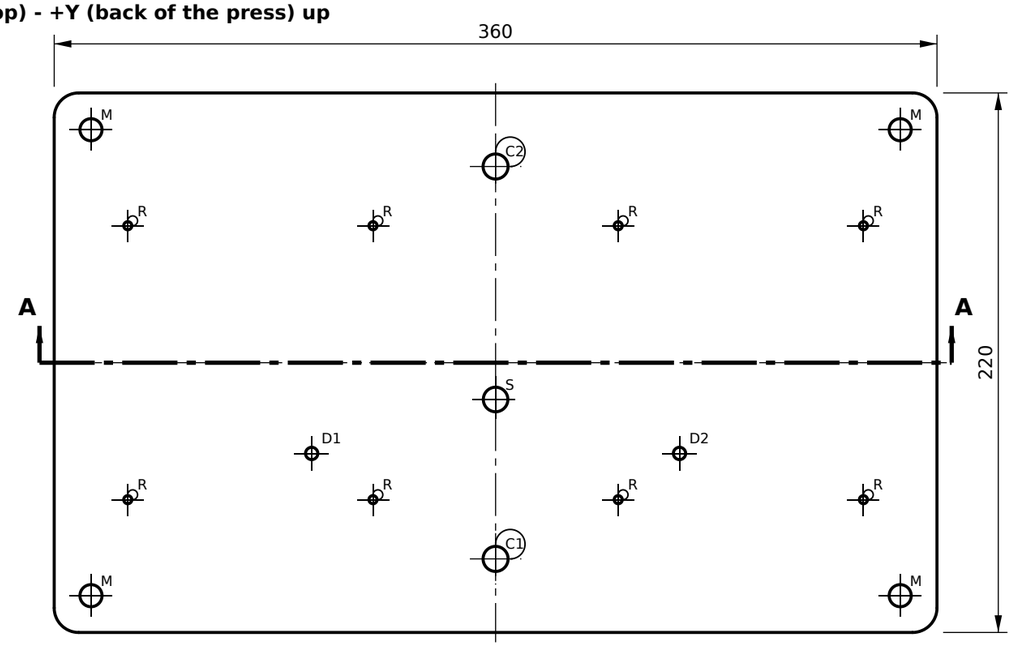
1. FRAME: screw the two columns L-104 into the base L-101 (M12, shoulder tight). Slide on the seat collars (top at Z 71.1), the return springs, the ram L-106 with its pressed-in bushes, then the up-stop collars (underside at Z 166.6).
2. HEAD: dowel and screw the two cheeks L-107 (stub pins L-111 pressed in) to the head L-105. Fit the head on the columns, washers and M12 nuts.
3. TOGGLE: clevis L-108 on the ram (2 dowels, 2 x M8 from below). Link L-110 in the clevis with pin B. Lever plates L-109 on the stub pins with the link between them; press pin A through plates and link (pin B likewise through the clevis); spacer tubes L-113 and grip spacer L-114 with 4 x M8 bolts.
4. TABLE: rails L-102 loosely on the base; stop pin in; table L-103 with ball plunger and knobs between the rails; push the rails to 0.05 a side and tighten.
5. STATION A: pedestal L-202 to the base block L-201 (2 dowels, 4 x M5 from below); springs, ring A L-203 on 4 shoulder screws from below; block to the table at x = -75 (2 dowels, 4 x M6 from below).
6. STATION B: cup ring L-205 on the riser L-204 (2 dowels, 4 x M5 countersunk); riser to the table at x = +75.
7. UPPER DIE: plug L-208 with the 0.4 shim pack L-209 to the die body L-206 (2 dowels, 4 x M6). Springs, upper ring L-207 on 4 shoulder screws, ball plungers set 0.4 proud. Die body to the ram (2 dowels, 4 x M8 from the top).
8. SET: station A under the ram, handle down: adjust the two stop bolts until the plug just touches a blank stack on the pedestal. Station B: add or remove shims until a test magnet comes out with its face level with the cup ring top.
9. USE: (1) station A out at the left: lay the tin in ring A's small pocket, photo face-up on it, film on top. (2) Slide to station A, pull the handle to the stop: the shell stays in the upper ring. (3) Station B out at the right: card with magnet, magnet down, into the cup. (4) Slide to station B, pull to the stop. (5) Slide back and lift the magnet out of the cup.

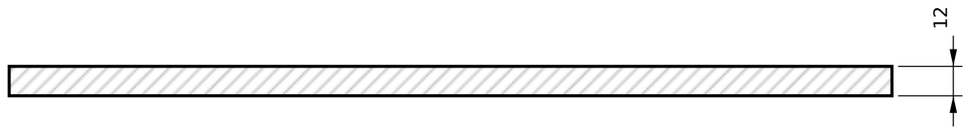
NOTES

1. Torques (8.8 screws, oiled): M4 3 Nm, M5 6 Nm, M6 10 Nm, M8 25 Nm, M12 column nuts 60 Nm. Thread-lock the shoulder screws and the stop-bolt jam nuts.
2. Before the first magnet: run both stations slowly with no material and check with a 0.1 feeler that the upper ring clears the pedestal all round (station A) and lands square on the cup ring (station B).

PROJECT: MagPress L1 · light photo-magnet press		REVISION: **L1-A**	SHEET: 5 / 28	
TITLE: **ASSEMBLY SEQUENCE AND USE**				
DRAWING NO.: **L-004**	QTY: -	SCALE: -	UNITS: mm	
MATERIAL: -				
HEAT TREATMENT / FINISH: -				
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-29	DRAWN	CHECKED

PLAN (top) - +Y (back of the press) up





SECTION A-A

FEATURES (positions from the part origin, mm)

KEY	POSITION	FEATURE
C	(0, +-80)	M12 THRU, square to the top face within 0.05 over 12 (columns L-104)
R	(+-50, +-150; Y +-55.85)	M4 x 10 (rails L-102), 8 off
S	(0, -15)	D10 H7 x 10.5 deep (stop pin D10 m6 x 16, 5.5 proud)
D	(+-75, -37)	Dimple: 90 deg spot drill to D5 (detent ball of the table)
M	(+-165, +-95)	D9 THRU (bench bolts M8)

DXF: L-101 profile (layer OUTLINE), holes (HOLES / TAP_DRILL / COUNTERBORE / POCKET).

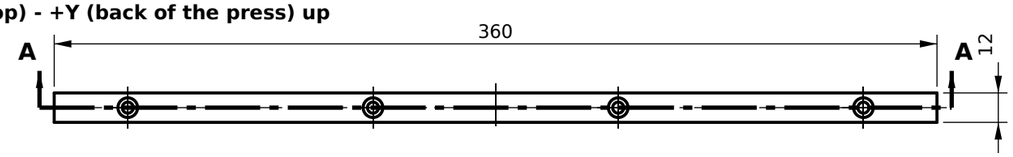
Mass about 7.39 kg each.

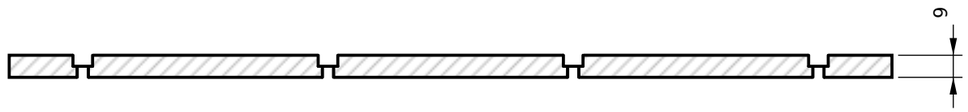
NOTES

1. Top face flat within 0.1 over the table slide area.
2. General tolerances ISO 2768-mK unless stated. Break sharp edges 0.3 x 45 deg, except the forming edges (stated).
3. Light design L1, one size (63.5 x 63.5). Material sizes are ASSUMED until a seller's pre-cut set is measured: parts marked MAT may change by a few tenths of a mm before manufacture.

PROJECT: MagPress L1 · light photo-magnet press			REVISION: **L1-A**	SHEET: 6 / 28
TITLE: **BASE PLATE**				
DRAWING NO.: **L-101**	QTY: 1	SCALE: 1:2.5	UNITS: mm	
MATERIAL: S275 (IS 2062 E250/E350) steel plate 12				
HEAT TREATMENT / FINISH: Black powder coat; slide area X +-170 x Y +-50 left bare, ground or surface-finished Ra 1.6, oiled				
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-29	DRAWN	CHECKED

PLAN (top) - +Y (back of the press) up





SECTION A-A

FEATURES (positions from the part origin, mm)

KEY	POSITION	FEATURE
-	(+-50, +-150; 0)	D4.5 THRU, c'bore D8 x 4.5 from the top (M4 x 10 cap screws)

DXF: L-102 profile (layer OUTLINE), holes (HOLES / TAP_DRILL / COUNTERBORE / POCKET).

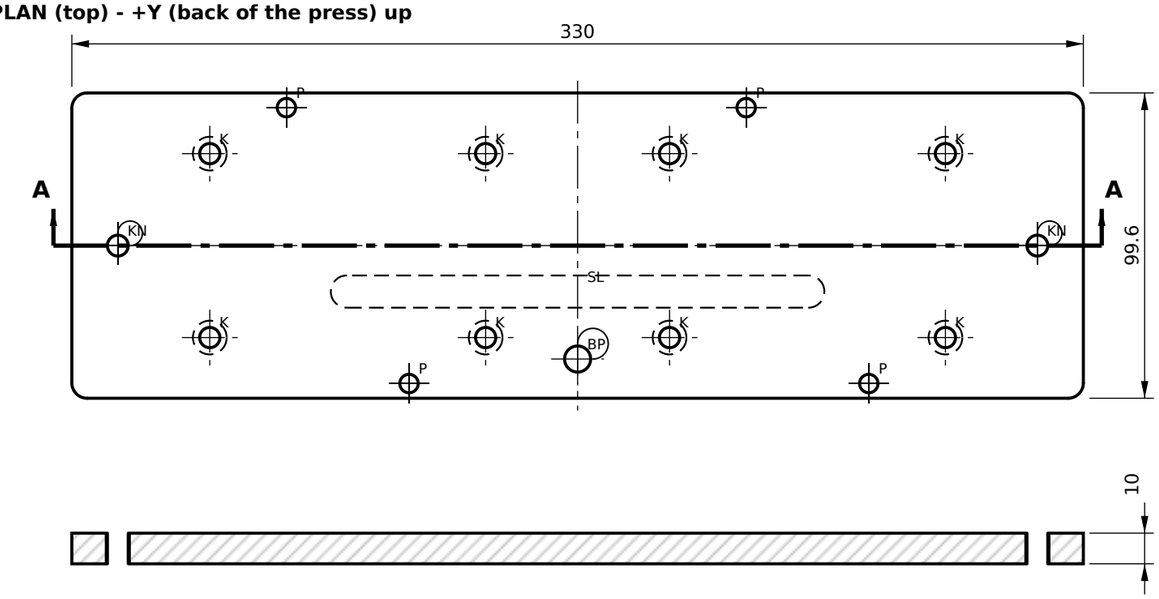
Mass about 0.30 kg each.

NOTES

1. Inner faces straight within 0.05 over the length. Set on assembly with the table between them: 0.05 a side, then tighten.
2. General tolerances ISO 2768-mK unless stated. Break sharp edges 0.3 x 45 deg, except the forming edges (stated).
3. Light design L1, one size (63.5 x 63.5). Material sizes are ASSUMED until a seller's pre-cut set is measured: parts marked MAT may change by a few tenths of a mm before manufacture.

PROJECT: MagPress L1 · light photo-magnet press		REVISION: **L1-A**	SHEET: 7 / 28
TITLE: **TABLE RAIL**			
DRAWING NO.: **L-102**	QTY: 2	SCALE: 1:2.5	UNITS: mm
MATERIAL: Bright mild steel flat 12 x 9 (EN8 / C45 bright bar)			
HEAT TREATMENT / FINISH: Black oxide or zinc; inner face oiled			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-29	DRAWN / CHECKED

PLAN (top) - +Y (back of the press) up



SECTION A-A

FEATURES (positions from the part origin, mm)

KEY	POSITION	FEATURE
K	station x +-75 +-45, y +-30	D6.6 THRU, c'bore D11 x 6.5 from the BOTTOM (M6 x 12, stations), 8 off
P	(-95, +45) (-55, -45) (55, +45) (95, -45)	D6 H7 THRU (dowels D6 m6 x 16 into the station blocks)
SL	y -15, x +-80.50	Slot 10.5 wide x 6 deep in the BOTTOM face, ends R5.25 (stop pin)
BP	(0, -37)	M10 THRU (ball plunger M10, ball 1.5 below the bottom face)
KN	(+-150, 0)	M8 THRU (DIN 319 knob D32)

DXF: L-103 profile (layer OUTLINE), holes (HOLES / TAP_DRILL / COUNTERBORE / POCKET).

Mass about 0.84 kg each.

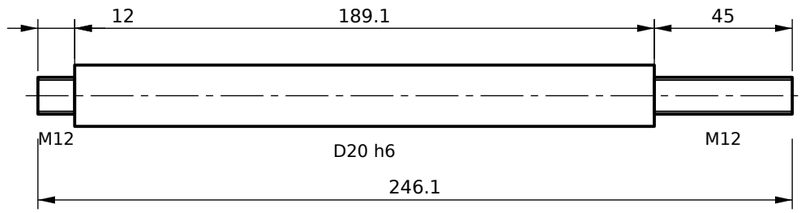
NOTES

1. Station A at x = -75, station B at x = +75. Bottom face flat within 0.05 (it slides on the base plate).
2. Pattern K/P is the same at both stations and matches L-201 and L-204.
3. General tolerances ISO 2768-mK unless stated. Break sharp edges 0.3 x 45 deg, except the forming edges (stated).
4. Light design L1, one size (63.5 x 63.5). Material sizes are ASSUMED until a seller's pre-cut set is measured: parts marked MAT may change by a few tenths of a mm before manufacture.

PROJECT: MagPress L1 · light photo-magnet press		REVISION: **L1-A**	SHEET: 8 / 28
TITLE: **SLIDING TABLE**			
DRAWING NO.: **L-103**	QTY: 1	SCALE: 1:2	UNITS: mm
MATERIAL: 6061-T6 aluminium plate 10			
HEAT TREATMENT / FINISH: Hard anodise (or clear anodise)			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-29	DRAWN / CHECKED

SIDE VIEW (axis horizontal)

Mass about 0.517 kg each.

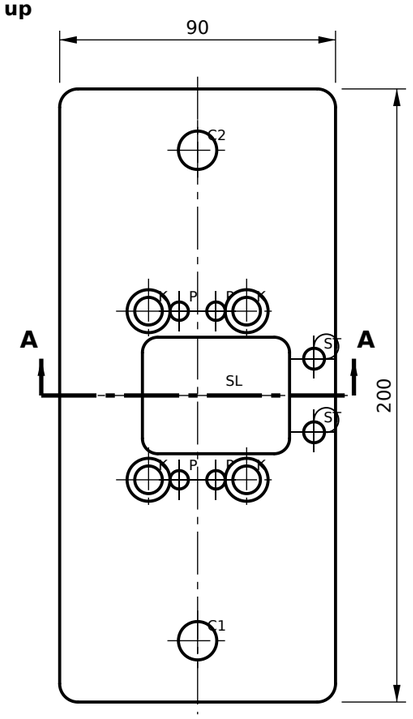


NOTES

1. Shoulders square to the axis within 0.02; the head (L-105) seats on the top shoulder.
2. Length between shoulders 189.1 sets the head height: tolerance +-0.05, both columns matched within 0.02.
3. General tolerances ISO 2768-mK unless stated. Break sharp edges 0.3 x 45 deg, except the forming edges (stated).
4. Light design L1, one size (63.5 x 63.5). Material sizes are ASSUMED until a seller's pre-cut set is measured: parts marked MAT may change by a few tenths of a mm before manufacture.

PROJECT: MagPress L1 · light photo-magnet press		REVISION: **L1-A**	SHEET: 9 / 28
TITLE: **COLUMN**			
DRAWING NO.: **L-104**	QTY: 2	SCALE: 1:2	UNITS: mm
MATERIAL: CK45 / C45 induction-hardened hard-chrome bar D20 h6			
HEAT TREATMENT / FINISH: Hard chrome (as bought); threads turned after cutting			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-29	DRAWN / CHECKED

PLAN (top) - +Y (back of the press) up



SECTION A-A

FEATURES (positions from the part origin, mm)

KEY	POSITION	FEATURE
C	(0, +-80)	D12.5 THRU (column threads); the head sits on the column shoulders
SL	x -18 ... +30, y +-19	Window THRU, corners R5 (lever crank, link, pin A and the clevis pass through)
K	(+-16, +-27.5)	D9 THRU, c'bore D14 x 8.5 from the BOTTOM (M8 x 35 cap screws into the cheeks)
P	(+-6, +-27.5)	D6 H7 x 8 from the TOP (cheek dowels D6 m6 x 16)
ST	(38, +-12)	M8 THRU (handle stops under the two lever plates: hex bolt M8 x 45 from below, jam nut under the head)

DXF: L-105 profile (layer OUTLINE), holes (HOLES / TAP_DRILL / COUNTERBORE / POCKET).

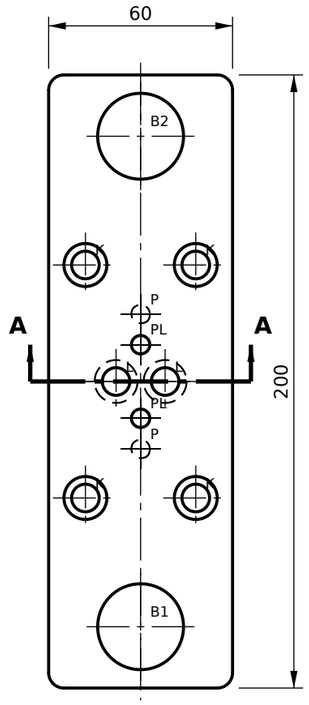
Mass about 3.63 kg each.

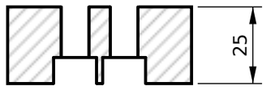
NOTES

1. Top face flat within 0.05 under the cheeks. Holes C at 160.00 +-0.05 and on the same line as K/P.
2. At the 15 kN abuse load: about 95 MPa in the section through the window (S275: 275 MPa yield).
3. General tolerances ISO 2768-mK unless stated. Break sharp edges 0.3 x 45 deg, except the forming edges (stated).
4. Light design L1, one size (63.5 x 63.5). Material sizes are ASSUMED until a seller's pre-cut set is measured: parts marked MAT may change by a few tenths of a mm before manufacture.

PROJECT: MagPress L1 · light photo-magnet press			REVISION: **L1-A**	SHEET: 10 / 28
TITLE: **HEAD**				
DRAWING NO.: **L-105**	QTY: 1	SCALE: 1:2	UNITS: mm	
MATERIAL: S275 (IS 2062 E250/E350) steel plate 30				
HEAT TREATMENT / FINISH: Black powder coat, holes masked				
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-29	DRAWN:	CHECKED:

PLAN (top) - +Y (back of the press) up





SECTION A-A

FEATURES (positions from the part origin, mm)

KEY	POSITION	FEATURE
B	(0, +-80)	D28 H7 THRU (bronze bushes D20 x D28 x 25, pressed in, ream D20 H7 after pressing)
K	(+-18, +-38)	D9 THRU, c'bore D14 x 8.5 from the TOP (M8 x 30 into the die body)
P	(0, +-22)	D6 H7 x 8 from the BOTTOM (die body dowels)
L	(+-8, 0)	D9 THRU, c'bore D14 x 8.5 from the BOTTOM (M8 x 25 into the clevis)
PL	(0, +-12)	D6 H7 x 8 from the TOP (clevis dowels)

DXF: L-106 profile (layer OUTLINE), holes (HOLES / TAP_DRILL / COUNTERBORE / POCKET).

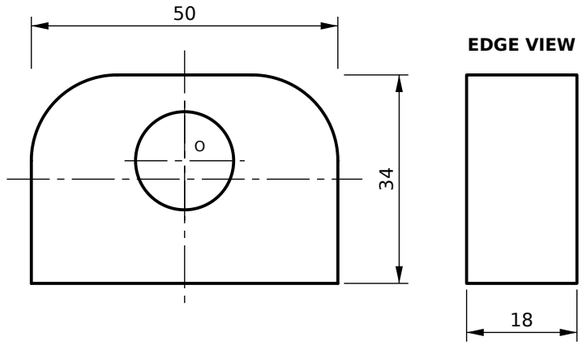
Mass about 1.99 kg each.

NOTES

1. Bores B: centres 160.00 +-0.03, square to the faces within 0.02 (they guide the ram on the columns).
2. Top and bottom faces parallel within 0.03.
3. General tolerances ISO 2768-mK unless stated. Break sharp edges 0.3 x 45 deg, except the forming edges (stated).
4. Light design L1, one size (63.5 x 63.5). Material sizes are ASSUMED until a seller's pre-cut set is measured: parts marked MAT may change by a few tenths of a mm before manufacture.

PROJECT: MagPress L1 · light photo-magnet press			REVISION: **L1-A**	SHEET: 11 / 28
TITLE: **RAM**				
DRAWING NO.: **L-106**	QTY: 1	SCALE: 1:2	UNITS: mm	
MATERIAL: S275 (IS 2062 E250/E350) steel plate 25				
HEAT TREATMENT / FINISH: Black oxide; bores masked				
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-29	DRAWN:	CHECKED:

PROFILE (side view, as cut)



FEATURES (positions from the part origin, mm)

KEY	POSITION	FEATURE
O	(0, 20)	D16 N7 THRU: stub pin O (L-111) pressed in; both cheeks bored as a pair
K	bottom edge, x +-16	M8 x 16 deep into the bottom edge, on its centre line
P	bottom edge, x +-6	D6 H7 x 10 deep into the bottom edge, on its centre line (dowels)

DXF: L-107 profile (layer OUTLINE), holes (HOLES / TAP_DRILL / COUNTERBORE / POCKET).

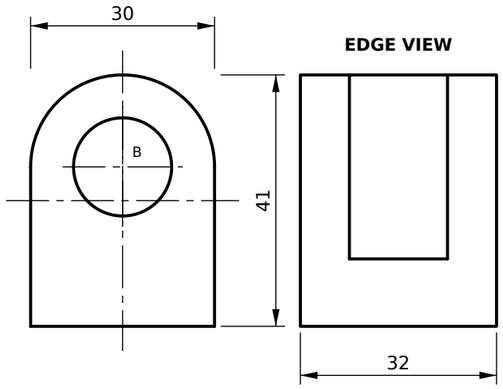
Mass about 0.18 kg each.

NOTES

1. Profile cut (laser / plasma), then mill the bottom edge flat and square to the bore.
2. Bore O and the bottom-edge holes with the two cheeks clamped as a mirrored pair.
3. General tolerances ISO 2768-mK unless stated. Break sharp edges 0.3 x 45 deg, except the forming edges (stated).
4. Light design L1, one size (63.5 x 63.5). Material sizes are ASSUMED until a seller's pre-cut set is measured: parts marked MAT may change by a few tenths of a mm before manufacture.

PROJECT: MagPress L1 · light photo-magnet press		REVISION: **L1-A**	SHEET: 12 / 28
TITLE: **PIVOT CHEEK**			
DRAWING NO.: **L-107**	QTY: 2	SCALE: 1:1	UNITS: mm
MATERIAL: S275 (IS 2062 E250/E350) steel plate 20, faced to 18 (or C45)			
HEAT TREATMENT / FINISH: Black oxide			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-29	DRAWN / CHECKED

PROFILE (side view, as cut)



FEATURES (positions from the part origin, mm)

KEY	POSITION	FEATURE
B	(0, 26)	D16 N7 THRU (pin B pressed in), across the slot
SL	centre	Slot 16 wide (Y), from Z 11 to the top, through in X (the link runs in it)
K	bottom face, x +-8, y 0	M8 x 10 deep from the bottom (2 off)
P	bottom face, x 0, y +-12	D6 H7 x 8 deep from the bottom (2 dowels)

DXF: L-108 profile (layer OUTLINE), holes (HOLES / TAP_DRILL / COUNTERBORE / POCKET).

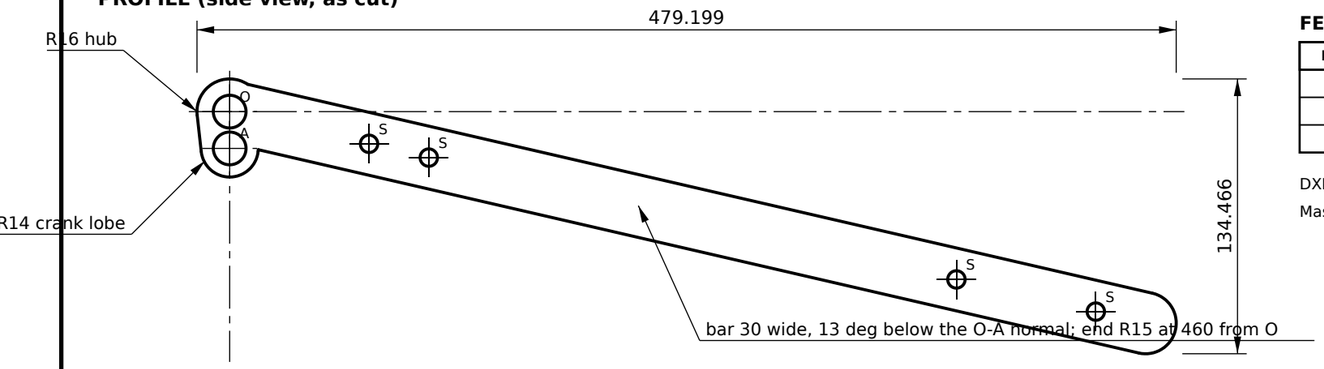
Mass about 0.15 kg each.

NOTES

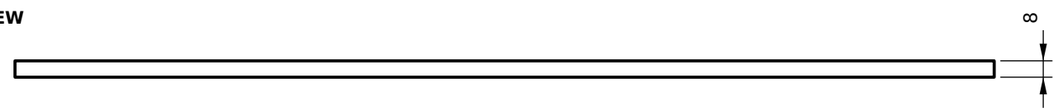
1. Shown: side profile (X-Z). Thickness 32 along Y = two 8 mm legs either side of the 16 mm slot.
2. Bottom face flat and square to the pin bore within 0.03.
3. General tolerances ISO 2768-mK unless stated. Break sharp edges 0.3 x 45 deg, except the forming edges (stated).
4. Light design L1, one size (63.5 x 63.5). Material sizes are ASSUMED until a seller's pre-cut set is measured: parts marked MAT may change by a few tenths of a mm before manufacture.

PROJECT: MagPress L1 · light photo-magnet press		REVISION: **L1-A**	SHEET: 13 / 28
TITLE: **RAM CLEVIS**			
DRAWING NO.: **L-108**	QTY: 1	SCALE: 1:1	UNITS: mm
MATERIAL: C45 / EN8 block 35 x 35 x 45			
HEAT TREATMENT / FINISH: Black oxide			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-29	DRAWN / CHECKED

PROFILE (side view, as cut)



EDGE VIEW



FEATURES (positions from the part origin, mm)

KEY	POSITION	FEATURE
O	(0, 0)	D16 H8 THRU (turns on the stub pin O)
A	(0, -18)	D16 N7 THRU (pin A pressed in), centres 18.00 +-0.03 from O
S	bar axis: 70, 100, 365, 435	D8.5 THRU (M8 bolts through the spacers), distances from O

DXF: L-109 profile (layer OUTLINE), holes (HOLES / TAP_DRILL / COUNTERBORE / POCKET).

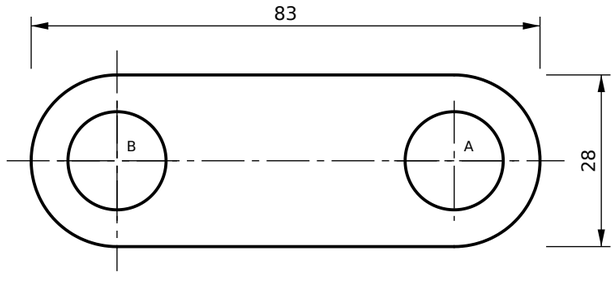
Mass about 0.90 kg each.

NOTES

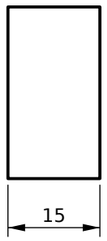
1. Two plates, a mirrored pair: profile cut together, ream O and A with the plates clamped as a pair.
2. Handle bar 30 wide at 77 deg to the crank line; grip centre 400 from O.
3. General tolerances ISO 2768-mK unless stated. Break sharp edges 0.3 x 45 deg, except the forming edges (stated).
4. Light design L1, one size (63.5 x 63.5). Material sizes are ASSUMED until a seller's pre-cut set is measured: parts marked MAT may change by a few tenths of a mm before manufacture.

PROJECT: MagPress L1 · light photo-magnet press		REVISION: **L1-A**	SHEET: 14 / 28
TITLE: **LEVER PLATE**			
DRAWING NO.: **L-109**	QTY: 2	SCALE: 1:3	UNITS: mm
MATERIAL: S275 (IS 2062 E250/E350) steel plate 8			
HEAT TREATMENT / FINISH: Black powder coat; grip end bare			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-29	DRAWN / CHECKED

PROFILE (side view, as cut)



EDGE VIEW



FEATURES (positions from the part origin, mm)

KEY	POSITION	FEATURE
A, B	(55, 0), (0, 0)	D16 H8 THRU, centres 55.00 +-0.03

DXF: L-110 profile (layer OUTLINE), holes (HOLES / TAP_DRILL / COUNTERBORE / POCKET).

Mass about 0.21 kg each.

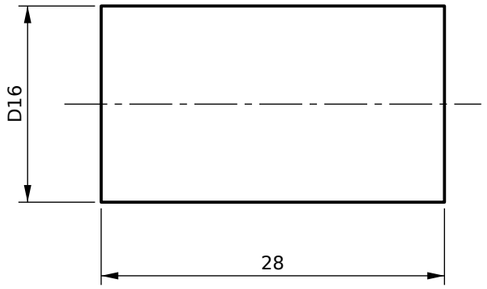
NOTES

1. Ends R14. Runs between the two lever plates (pin A) and in the clevis slot (pin B).
2. General tolerances ISO 2768-mK unless stated. Break sharp edges 0.3 x 45 deg, except the forming edges (stated).
3. Light design L1, one size (63.5 x 63.5). Material sizes are ASSUMED until a seller's pre-cut set is measured: parts marked MAT may change by a few tenths of a mm before manufacture.

PROJECT: MagPress L1 · light photo-magnet press		REVISION: **L1-A**	SHEET: 15 / 28
TITLE: **TOGGLE LINK**			
DRAWING NO.: **L-110**	QTY: 1	SCALE: 1:1	UNITS: mm
MATERIAL: S355 (IS 2062 E350) plate 16, faced to 15			
HEAT TREATMENT / FINISH: Black oxide			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-29	DRAWN / CHECKED

SIDE VIEW (2:1)

Mass about 0.044 kg each.



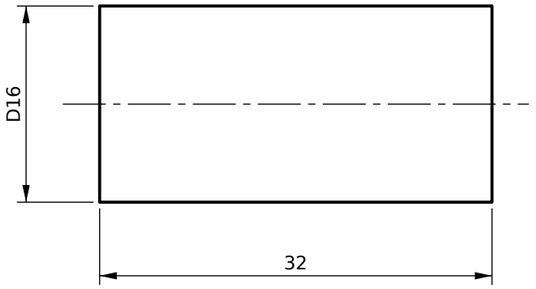
NOTES

1. Pressed into the cheek (D16 N7) flush with its outer face; the lever plate turns on the inner 7.5 mm, a 2.5 mm gap between plate and cheek. Grease.
2. General tolerances ISO 2768-mK unless stated. Break sharp edges 0.3 x 45 deg, except the forming edges (stated).
3. Light design L1, one size (63.5 x 63.5). Material sizes are ASSUMED until a seller's pre-cut set is measured: parts marked MAT may change by a few tenths of a mm before manufacture.

PROJECT: MagPress L1 · light photo-magnet press		REVISION: **L1-A**	SHEET: 16 / 28
TITLE: **PIN O (STUB)**			
DRAWING NO.: **L-111**	QTY: 2	SCALE: 1:1	UNITS: mm
MATERIAL: EN8 / C45 bright bar (or EN19 for longer life)			
HEAT TREATMENT / FINISH: Ground D16 m6; chamfer 1 x 45 both ends			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-29	DRAWN / CHECKED

SIDE VIEW (2:1)

Mass about 0.051 kg each.

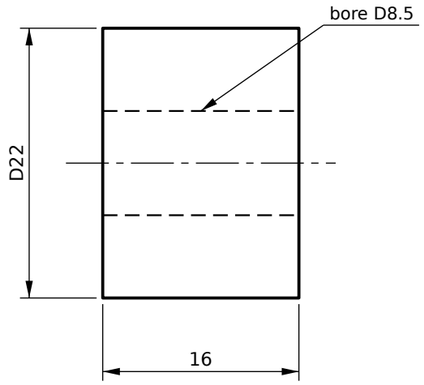


NOTES

1. Pressed into the lever plates (pin A) or the clevis legs (pin B), flush both sides; the link turns on it (link bores H8). No rings: a ring on pin A would hit the stub pins O as the crank turns. Grease the link bores.
2. General tolerances ISO 2768-mK unless stated. Break sharp edges 0.3 x 45 deg, except the forming edges (stated).
3. Light design L1, one size (63.5 x 63.5). Material sizes are ASSUMED until a seller's pre-cut set is measured: parts marked MAT may change by a few tenths of a mm before manufacture.

PROJECT: MagPress L1 · light photo-magnet press		REVISION: **L1-A**	SHEET: 17 / 28
TITLE: **PIN A / PIN B**			
DRAWING NO.: **L-112**	QTY: 2	SCALE: 1:1	UNITS: mm
MATERIAL: EN8 / C45 bright bar (or EN19)			
HEAT TREATMENT / FINISH: Ground D16 m6; chamfer 1 x 45 both ends			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-29	DRAWN / CHECKED

SIDE VIEW (2:1)



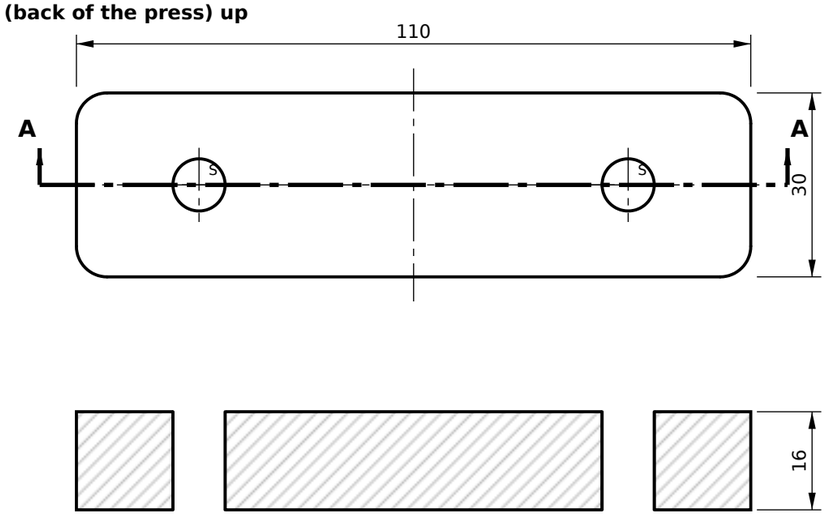
Mass about 0.041 kg each.

NOTES

1. Length 16.00 +0.05/0: sets the gap between the lever plates.
2. General tolerances ISO 2768-mK unless stated. Break sharp edges 0.3 x 45 deg, except the forming edges (stated).
3. Light design L1, one size (63.5 x 63.5). Material sizes are ASSUMED until a seller's pre-cut set is measured: parts marked MAT may change by a few tenths of a mm before manufacture.

PROJECT: MagPress L1 · light photo-magnet press		REVISION: **L1-A**	SHEET: 18 / 28
TITLE: **LEVER SPACER TUBE**			
DRAWING NO.: **L-113**	QTY: 2	SCALE: 1:1	UNITS: mm
MATERIAL: EN8 / mild steel round bar D22			
HEAT TREATMENT / FINISH: Zinc or black oxide			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-29	DRAWN / CHECKED

PLAN (top) - +Y (back of the press) up



SECTION A-A

FEATURES (positions from the part origin, mm)

KEY	POSITION	FEATURE
S	(+-35, 0)	D8.5 THRU (M8 x 35 bolts through both lever plates)

DXF: L-114 profile (layer OUTLINE), holes (HOLES / TAP_DRILL / COUNTERBORE / POCKET).

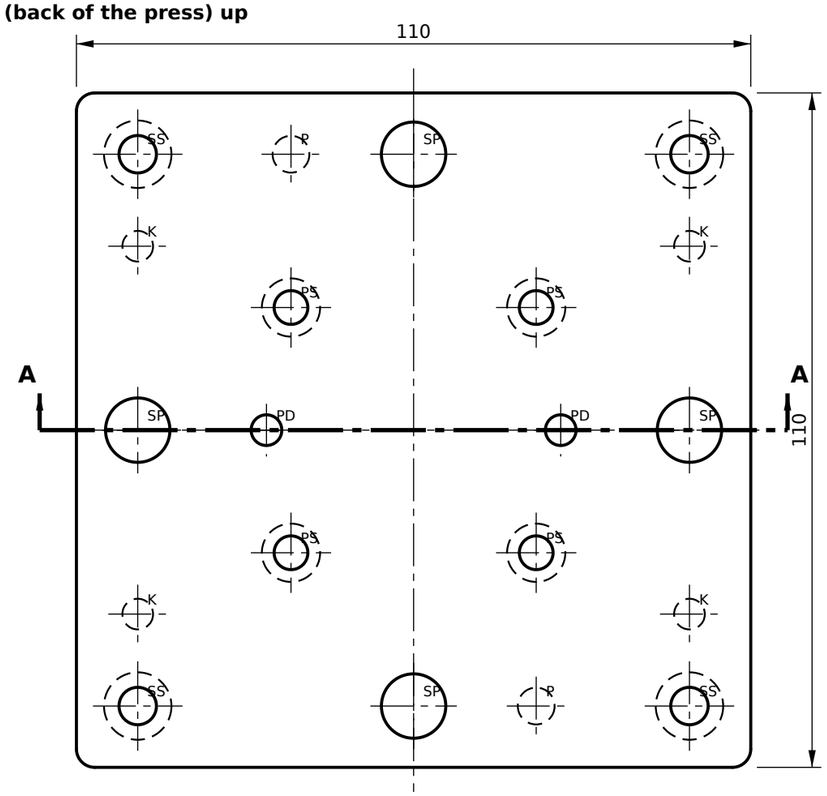
Mass about 0.40 kg each.

NOTES

1. Fills the grip end of the lever (32 x 30 with the plates); fit the slip-on grip over it.
2. General tolerances ISO 2768-mK unless stated. Break sharp edges 0.3 x 45 deg, except the forming edges (stated).
3. Light design L1, one size (63.5 x 63.5). Material sizes are ASSUMED until a seller's pre-cut set is measured: parts marked MAT may change by a few tenths of a mm before manufacture.

PROJECT: MagPress L1 · light photo-magnet press		REVISION: **L1-A**	SHEET: 19 / 28
TITLE: **GRIP SPACER**			
DRAWING NO.: **L-114**	QTY: 1	SCALE: 1:1	UNITS: mm
MATERIAL: S275 (IS 2062 E250/E350) steel plate 16			
HEAT TREATMENT / FINISH: Black powder coat			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-29	DRAWN / CHECKED

PLAN (top) - +Y (back of the press) up





SECTION A-A

FEATURES (positions from the part origin, mm)

KEY	POSITION	FEATURE
SP	(+-45, 0), (0, +-45)	D10.5 THRU (ring A springs ISO 10243 green D10 x 38; they seat on the table)
SS	(+-45, +-45)	D6.1 THRU (shoulder guide), c'bore D11 x 13.5 from the BOTTOM (ISO 7379 D6 x 16 / M5)
PS	(+-20, +-20)	D5.5 THRU, c'bore D9.5 x 5.5 from the BOTTOM (M5 x 16 into the pedestal)
PD	(+-24, 0)	D5 H7 x 6 from the TOP (pedestal dowels D5 m6 x 12)
K	(+-45, +-30)	M6 x 10 from the BOTTOM (table screws)
P	(-20, +45), (20, -45)	D6 H7 x 8 from the BOTTOM (table dowels)

DXF: L-201 profile (layer OUTLINE), holes (HOLES / TAP_DRILL / COUNTERBORE / POCKET).

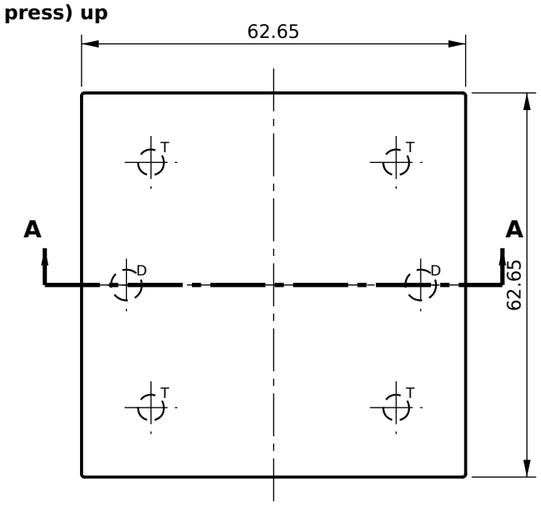
Mass about 0.64 kg each.

NOTES

1. Top face (ring A stop) parallel to the bottom within 0.02; thickness 21 +-0.02 (sets the stroke-1 depth).
2. General tolerances ISO 2768-mK unless stated. Break sharp edges 0.3 x 45 deg, except the forming edges (stated).
3. Light design L1, one size (63.5 x 63.5). Material sizes are ASSUMED until a seller's pre-cut set is measured: parts marked MAT may change by a few tenths of a mm before manufacture.

PROJECT: MagPress L1 · light photo-magnet press			REVISION: **L1-A**	SHEET: 20 / 28
TITLE: **STATION A BASE BLOCK**				
DRAWING NO.: **L-201**	QTY: 1	SCALE: 1:1	UNITS: mm	
MATERIAL: 6061-T6 aluminium plate 16, faced to 21				
HEAT TREATMENT / FINISH: Clear anodise				
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-29	DRAWN:	CHECKED:

PLAN (top) - +Y (back of the press) up





SECTION A-A

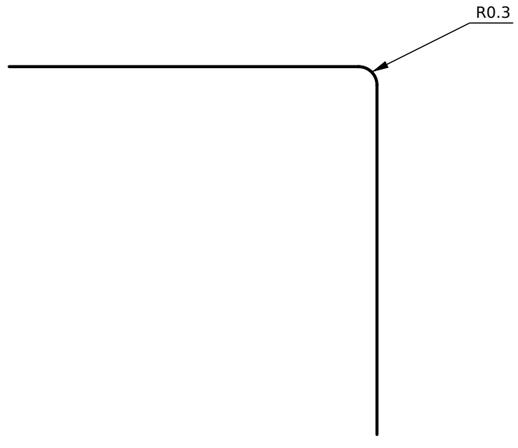
FEATURES (positions from the part origin, mm)

KEY	POSITION	FEATURE
T	(+-20, +-20)	M5 x 10 from the BOTTOM
D	(+-24, 0)	D5 H7 x 6 from the BOTTOM (dowels)
-	top edge	R0.3 all round the top edge (the tin folds over it); corners R0.5 in plan

DXF: L-202 profile (layer OUTLINE), holes (HOLES / TAP_DRILL / COUNTERBORE / POCKET).

Mass about 0.44 kg each.

DETAIL: pedestal top edge (10:1)

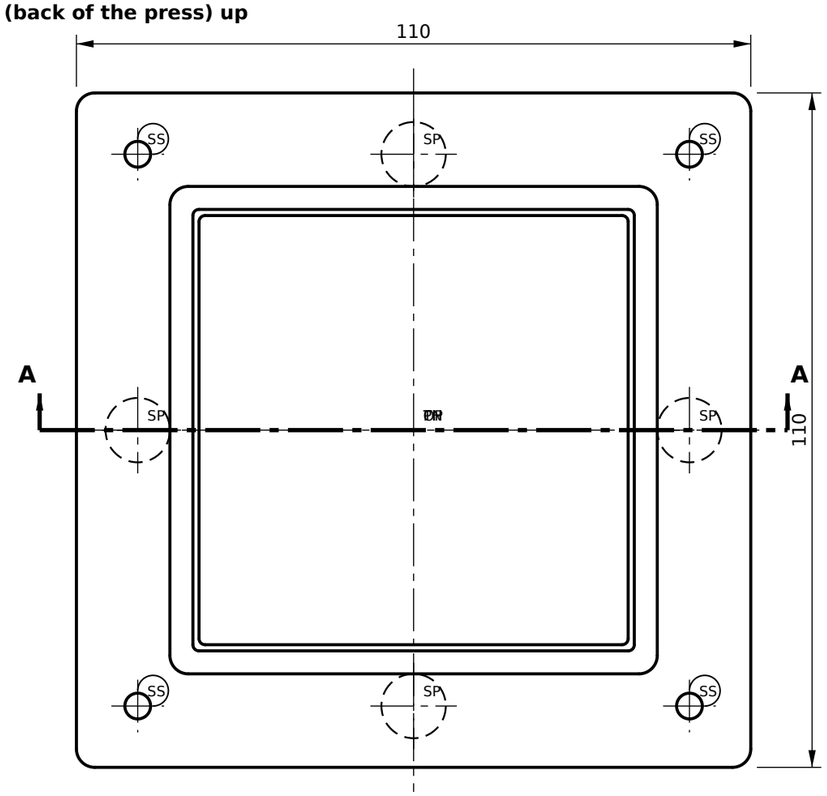


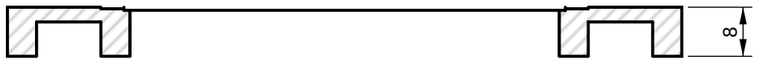
NOTES

1. MAT: 62.65 = face 63.5 - 2 x stack 0.425 (tin 0.23 + photo 0.12 + film 0.075).
2. Height 14.675 +-0.01 (sets the face level with the cup ring top).
3. Wire-EDM the outline after hardening (or grind); sides square to the top within 0.01.
4. General tolerances ISO 2768-mK unless stated. Break sharp edges 0.3 x 45 deg, except the forming edges (stated).
5. Light design L1, one size (63.5 x 63.5). Material sizes are ASSUMED until a seller's pre-cut set is measured: parts marked MAT may change by a few tenths of a mm before manufacture.

PROJECT: MagPress L1 · light photo-magnet press		REVISION: **L1-A**	SHEET: 21 / 28
TITLE: **PEDESTAL**			
DRAWING NO.: **L-202**	QTY: 1	SCALE: 1:1	UNITS: mm
MATERIAL: D2 / 1.2379 tool steel (prototype: 1.2312 P20 prehardened 30 HRC)			
HEAT TREATMENT / FINISH: Harden and temper 58-60 HRC (D2); prototype in P20: no heat treatment; top face and edge polished Ra 0.2			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-29	DRAWN / CHECKED

PLAN (top) - +Y (back of the press) up





SECTION A-A

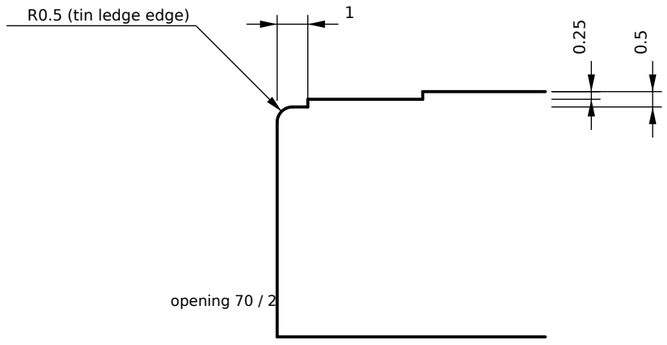
FEATURES (positions from the part origin, mm)

KEY	POSITION	FEATURE
OP	centre	70 x 70 THRU, corners R1; top edge R0.5 (tin ledge)
TN	centre	Tin pocket 72 x 72 x 0.5 deep from the top, corners R1
PH	centre	Photo / film pocket 79.5 x 79.5 x 0.25 deep from the top, corners R3
SS	(+-45, +-45)	M5 THRU (shoulder screws from below)
SP	(+-45, 0), (0, +-45)	D10.5 x 4 from the BOTTOM (spring seats)

DXF: L-203 profile (layer OUTLINE), holes (HOLES / TAP_DRILL / COUNTERBORE / POCKET).

Mass about 0.43 kg each.

DETAIL: opening edge and pockets, half section (5:1)

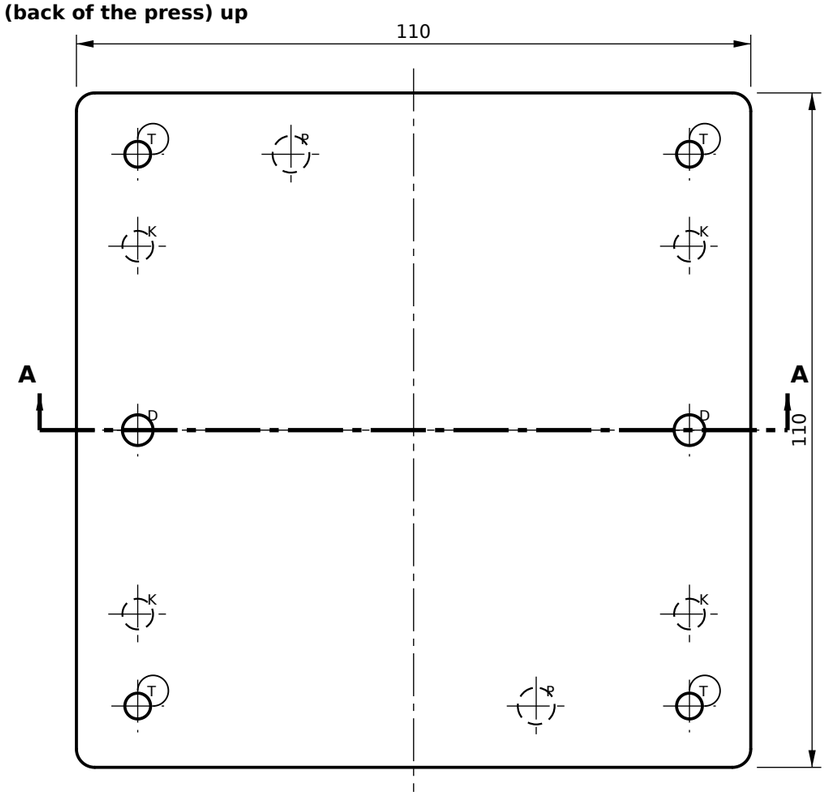


NOTES

1. MAT: pockets follow the tin and photo sizes (tin 71.5, photo 79); opening = tin - 1.5 (0.75 ledge a side).
2. Wire-EDM the opening after hardening; mill the pockets before hardening and finish to depth after.
3. General tolerances ISO 2768-mK unless stated. Break sharp edges 0.3 x 45 deg, except the forming edges (stated).
4. Light design L1, one size (63.5 x 63.5). Material sizes are ASSUMED until a seller's pre-cut set is measured: parts marked MAT may change by a few tenths of a mm before manufacture.

PROJECT: MagPress L1 · light photo-magnet press	REVISION: **L1-A**	SHEET: 22 / 28	
TITLE: **RING A (BLANK HOLDER)**			
DRAWING NO.: **L-203**	QTY: 1	SCALE: 1:1	UNITS: mm
MATERIAL: D2 / 1.2379 tool steel (prototype: 1.2312 P20 prehardened 30 HRC)			
HEAT TREATMENT / FINISH: Harden and temper 58-60 HRC (D2); prototype in P20: no heat treatment; opening edge polished			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-29	DRAWN / CHECKED

PLAN (top) - +Y (back of the press) up





SECTION A-A

FEATURES (positions from the part origin, mm)

KEY	POSITION	FEATURE
K	(+-45, +-30)	M6 x 12 from the BOTTOM (table screws)
P	(-20, +45), (20, -45)	D6 H7 x 8 from the BOTTOM
T	(+-45, +-45)	M5 x 10 from the TOP (cup ring screws)
D	(+-45, 0)	D5 H7 x 6 from the TOP (cup ring dowels)

DXF: L-204 profile (layer OUTLINE), holes (HOLES / TAP_DRILL / COUNTERBORE / POCKET).

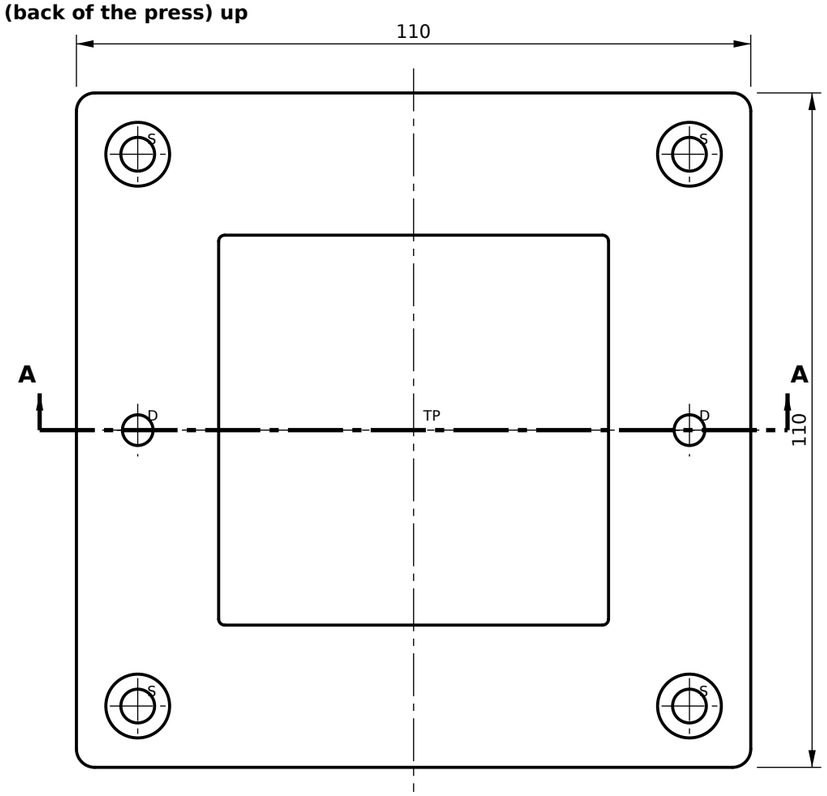
Mass about 1.03 kg each.

NOTES

1. MAT: thickness 31.675 +-0.02 = face level 50.0 - cup ring 4.425 - table top 22. Top face flat within 0.01.
2. Aluminium so the magnet sheet does not stick to the floor.
3. General tolerances ISO 2768-mK unless stated. Break sharp edges 0.3 x 45 deg, except the forming edges (stated).
4. Light design L1, one size (63.5 x 63.5). Material sizes are ASSUMED until a seller's pre-cut set is measured: parts marked MAT may change by a few tenths of a mm before manufacture.

PROJECT: MagPress L1 · light photo-magnet press		REVISION: **L1-A**	SHEET: 23 / 28
TITLE: **STATION B RISER**			
DRAWING NO.: **L-204**	QTY: 1	SCALE: 1:1	UNITS: mm
MATERIAL: 6061-T6 aluminium plate 25, faced to 31.675			
HEAT TREATMENT / FINISH: Hard anodise (the magnet sits on the top face)			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-29	DRAWN / CHECKED

PLAN (top) - +Y (back of the press) up





SECTION A-A

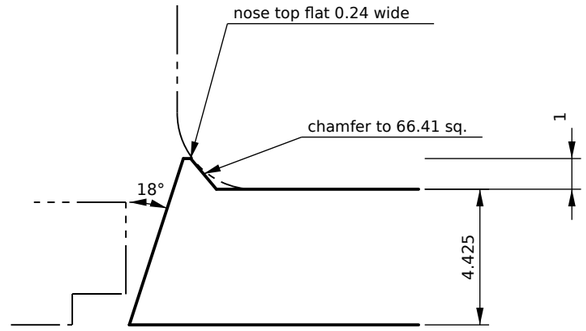
FEATURES (positions from the part origin, mm)

KEY	POSITION	FEATURE
TP	centre	Opening THRU, 18 deg taper: 64.25 square at the nose top, 63.6 at the landing face, 60.72 at the bottom; corners R
N	round TP	Nose 1 high above the landing face: flat top to 64.73 square, chamfer down to 66.41 square (see detail)
LF	outside N	Landing face (the upper ring lands here): 1 below the nose top, 4.425 above the bottom face
S	(+-45, +-45)	D5.5 THRU, 90 deg csk D10.4 from the landing face (M5 x 12 csk)
D	(+-45, 0)	D5 H7 THRU (dowels D5 m6 x 10)

DXF: L-205 profile (layer OUTLINE), holes (HOLES / TAP_DRILL / COUNTERBORE / POCKET).

Mass about 0.27 kg each.

DETAIL: taper and nose, half section; upper ring landed (phantom) (5:1)

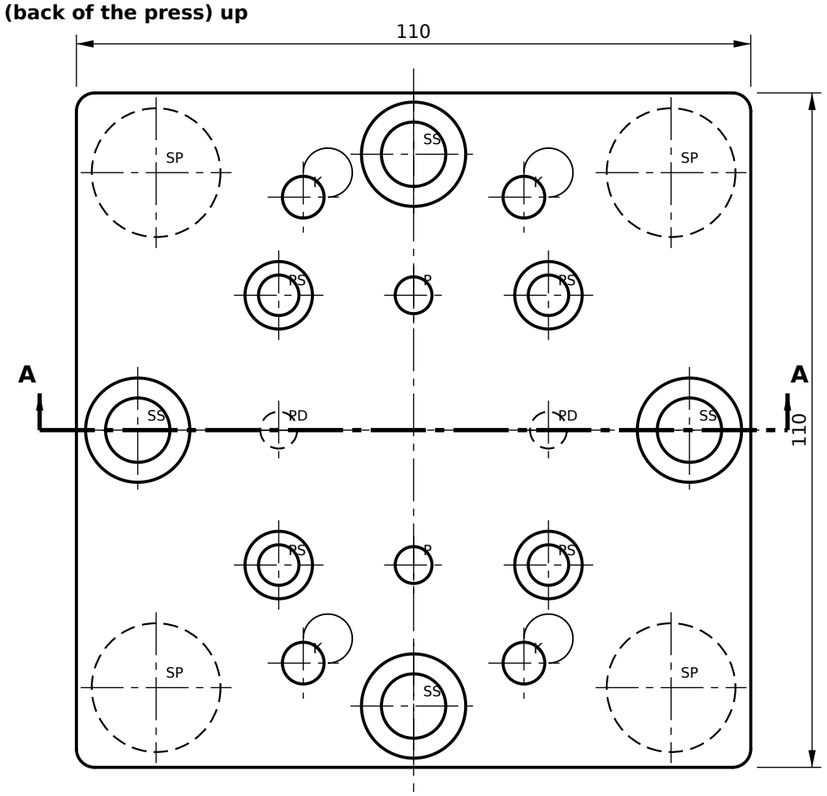


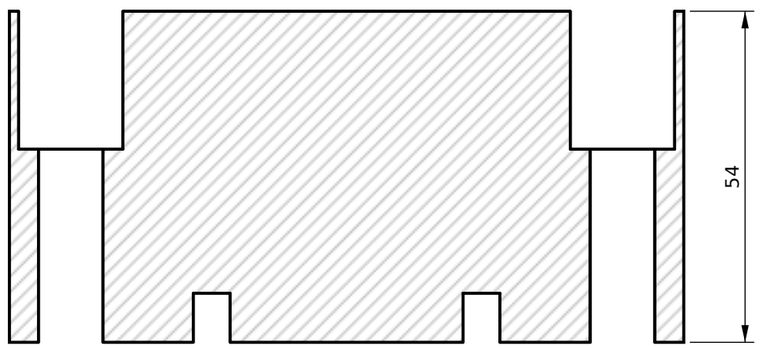
NOTES

1. MAT: landing face 4.425 above the bottom = finished magnet (stack 0.425 + acrylic 3 + magnet 1); grind the bottom and the landing face.
2. Mill the landing face 1.0 down round the nose and the nose chamfer before hardening; taper THRU by 4-axis wire-EDM after hardening. The nose fills the upper ring's mouth (0.05 clear) so the shell wall stays supported; the taper turns the flap in onto the card.
3. General tolerances ISO 2768-mK unless stated. Break sharp edges 0.3 x 45 deg, except the forming edges (stated).
4. Light design L1, one size (63.5 x 63.5). Material sizes are ASSUMED until a seller's pre-cut set is measured: parts marked MAT may change by a few tenths of a mm before manufacture.

PROJECT: MagPress L1 · light photo-magnet press	REVISION: **L1-A**	SHEET: 24 / 28	
TITLE: **CUP RING (TAPER + NOSE)**			
DRAWING NO.: **L-205**	QTY: 1	SCALE: 1:1	UNITS: mm
MATERIAL: D2 / 1.2379 tool steel (prototype: 1.2312 P20 prehardened 30 HRC)			
HEAT TREATMENT / FINISH: Harden and temper 58-60 HRC (D2); prototype in P20: no heat treatment; taper and nose polished Ra 0.2			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-29	DRAWN / CHECKED

PLAN (top) - +Y (back of the press) up





SECTION A-A

FEATURES (positions from the part origin, mm)

KEY	POSITION	FEATURE
SP	(+-42, +-42)	D21 x 48 from the BOTTOM (upper springs ISO 10243 red D20 x 76)
SS	(+-45, 0), (0, +-45)	D10.5 THRU, c'bore D17 x 22.5 from the TOP (ISO 7379 D10 x 40 / M8)
PS	(+-22, +-22)	D6.6 THRU, c'bore D11 x 6.5 from the TOP (M6 x 55 into the plug)
PD	(+-22, 0)	D6 H7 x 8 from the BOTTOM (plug dowels D6 m6 x 16)
K	(+-18, +-38)	M8 x 16 from the TOP (ram screws)
P	(0, +-22)	D6 H7 x 8 from the TOP (ram dowels)

DXF: L-206 profile (layer OUTLINE), holes (HOLES / TAP_DRILL / COUNTERBORE / POCKET).

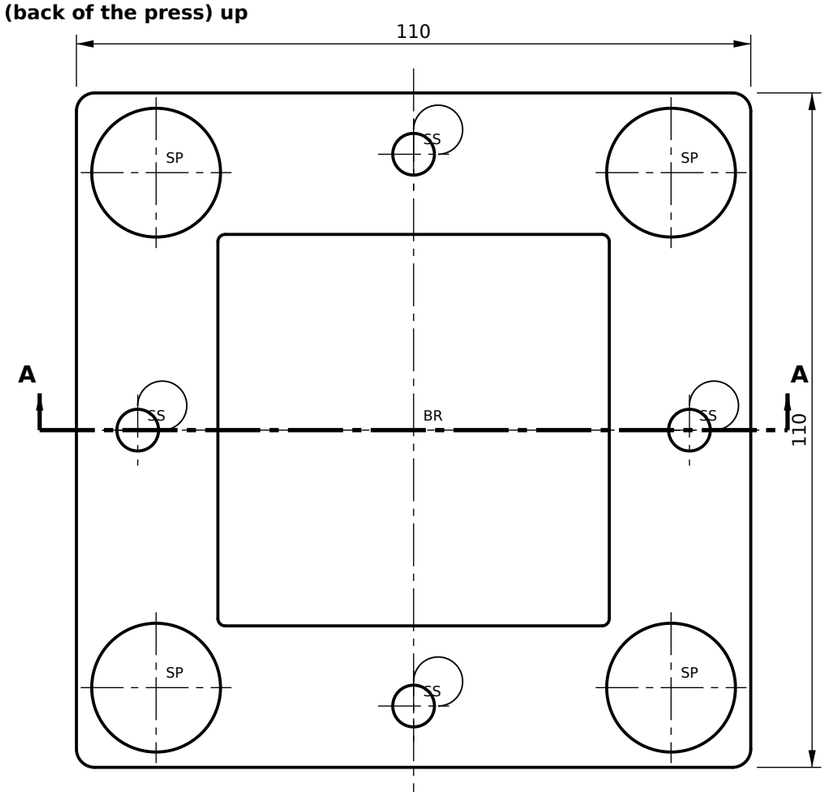
Mass about 1.46 kg each.

NOTES

1. Faces parallel within 0.02. The shoulder screws guide the upper ring: holes SS square to the faces within 0.02.
2. General tolerances ISO 2768-mK unless stated. Break sharp edges 0.3 x 45 deg, except the forming edges (stated).
3. Light design L1, one size (63.5 x 63.5). Material sizes are ASSUMED until a seller's pre-cut set is measured: parts marked MAT may change by a few tenths of a mm before manufacture.

PROJECT: MagPress L1 · light photo-magnet press		REVISION: **L1-A**	SHEET: 25 / 28
TITLE: **UPPER DIE BODY**			
DRAWING NO.: **L-206**	QTY: 1	SCALE: 1:1	UNITS: mm
MATERIAL: 6061-T6 aluminium plate 40, faced to 54			
HEAT TREATMENT / FINISH: Clear anodise			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-29	DRAWN / CHECKED

PLAN (top) - +Y (back of the press) up





SECTION A-A

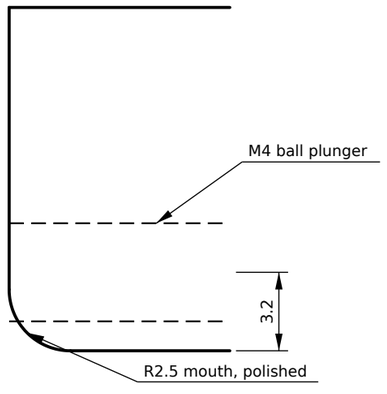
FEATURES (positions from the part origin, mm)

KEY	POSITION	FEATURE
BR	centre	Bore 63.85 x 63.85 THRU, corners R1.2; bottom edge R2.5 (mouth)
SP	(+-42, +-42)	D21 x 8 from the TOP (spring seats)
SS	(+-45, 0), (0, +-45)	M8 THRU (shoulder screws)
BP	side faces, 3.2 up	M4 THRU to the bore, 4 off: +X face y +12, -X face y -12, +Y face x -12, -Y face x +12
		(ball plungers M4 set 0.4 proud in the bore, locked by an M4 set screw behind)

DXF: L-207 profile (layer OUTLINE), holes (HOLES / TAP_DRILL / COUNTERBORE / POCKET).

Mass about 0.76 kg each.

DETAIL: bore mouth, half section (4:1)

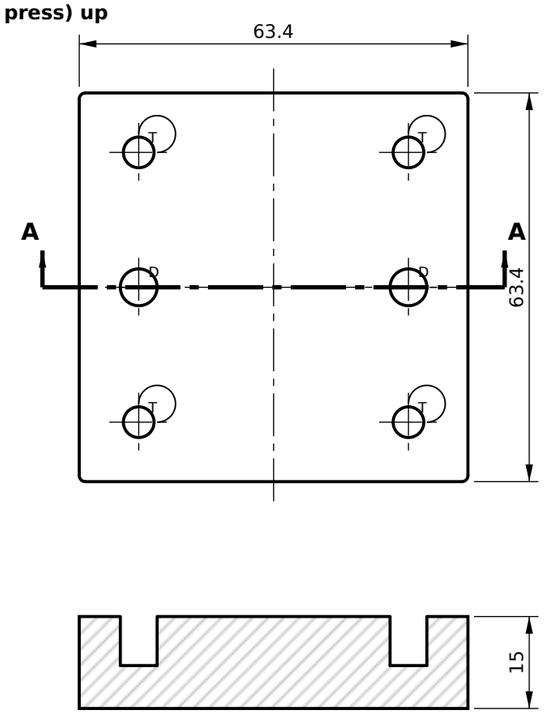


NOTES

1. MAT: bore = face 63.5 + 2 x 0.175 (the shell wall stands inside it); mouth R2.5 lets the flat tin slide in without being drawn.
2. Wire-EDM the bore after hardening; grind the mouth radius blend. Drill and tap BP before hardening.
3. General tolerances ISO 2768-mK unless stated. Break sharp edges 0.3 x 45 deg, except the forming edges (stated).
4. Light design L1, one size (63.5 x 63.5). Material sizes are ASSUMED until a seller's pre-cut set is measured: parts marked MAT may change by a few tenths of a mm before manufacture.

PROJECT: MagPress L1 · light photo-magnet press		REVISION: **L1-A**	SHEET: 26 / 28
TITLE: **UPPER RING**			
DRAWING NO.: **L-207**	QTY: 1	SCALE: 1:1	UNITS: mm
MATERIAL: D2 / 1.2379 tool steel (prototype: 1.2312 P20 prehardened 30 HRC)			
HEAT TREATMENT / FINISH: Harden and temper 58-60 HRC (D2); prototype in P20: no heat treatment; bore and mouth polished Ra 0.2			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-29	DRAWN / CHECKED

PLAN (top) - +Y (back of the press) up



SECTION A-A

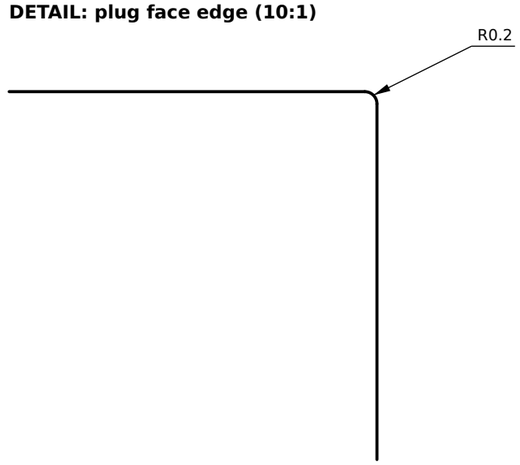
FEATURES (positions from the part origin, mm)

KEY	POSITION	FEATURE
T	(+-22, +-22)	M6 x 12 from the TOP
D	(+-22, 0)	D6 H7 x 8 from the TOP (dowels)
-	bottom edge	R0.2 all round

DXF: L-208 profile (layer OUTLINE), holes (HOLES / TAP_DRILL / COUNTERBORE / POCKET).

Mass about 0.46 kg each.

DETAIL: plug face edge (10:1)

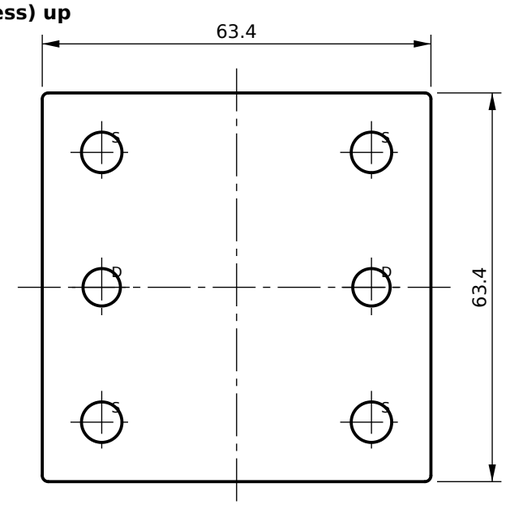


NOTES

1. 63.4: 0.05 a side inside the magnet face, so it passes inside the cup ring's nose; 0.225 a side clear of the upper ring bore (the ring is guided by its shoulder screws). Outline +-0.02 after hardening.
2. General tolerances ISO 2768-mK unless stated. Break sharp edges 0.3 x 45 deg, except the forming edges (stated).
3. Light design L1, one size (63.5 x 63.5). Material sizes are ASSUMED until a seller's pre-cut set is measured: parts marked MAT may change by a few tenths of a mm before manufacture.

PROJECT: MagPress L1 · light photo-magnet press		REVISION: **L1-A**	SHEET: 27 / 28
TITLE: **PLUG**			
DRAWING NO.: **L-208**	QTY: 1	SCALE: 1:1	UNITS: mm
MATERIAL: D2 / 1.2379 tool steel (prototype: 1.2312 P20 prehardened 30 HRC)			
HEAT TREATMENT / FINISH: Harden and temper 58-60 HRC (D2); prototype in P20: no heat treatment; face lapped Ra 0.1 (it presses the film side)			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-29	DRAWN / CHECKED

PLAN (top) - +Y (back of the press) up



FEATURES (positions from the part origin, mm)

KEY	POSITION	FEATURE
S	(+-22, +-22)	D6.6 THRU
D	(+-22, 0)	D6.1 THRU

DXF: L-209 profile (layer OUTLINE), holes (HOLES / TAP_DRILL / COUNTERBORE / POCKET).

Mass about 0.01 kg each.

NOTES

1. Nominal pack 0.4 between plug and die body: add or remove to set the plug face level with the cup ring top (station B) at the bottom of the stroke.
2. General tolerances ISO 2768-mK unless stated. Break sharp edges 0.3 x 45 deg, except the forming edges (stated).
3. Light design L1, one size (63.5 x 63.5). Material sizes are ASSUMED until a seller's pre-cut set is measured: parts marked MAT may change by a few tenths of a mm before manufacture.

PROJECT: MagPress L1 · light photo-magnet press			REVISION: **L1-A**	SHEET: 28 / 28
TITLE: **PLUG SHIM**				
DRAWING NO.: **L-209**	QTY: 2 x 0.1, 1 x 0.2	SCALE: 1:1	UNITS: mm	
MATERIAL: Stainless shim stock 0.1 / 0.2 (AISI 301 / 304)				
HEAT TREATMENT / FINISH: Deburr; flat				
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-29	DRAWN	CHECKED